

Data Cleaning & Visualization Report

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Dataset: Titanic (Seaborn)

1. Data Inspection & Initial State

Original Dataset Shape: 891 rows x 15 columns

Duplicate Rows Found: 107

Missing Values Summary:

Column	Null Count	Percentage
age	177	19.87%
embarked	2	0.22%
deck	688	77.22%
embark_town	2	0.22%

2. Descriptive Statistics

Feature	count	unique	top	mean	std	min	max
survived	891.0	-	-	0.38383 838383 83	0.48659 245426 48	0.0	1.0
pclass	891.0	-	-	2.30864 197530 86	0.83607 124097 70	1.0	3.0

Feature	count	unique	top	mean	std	min	max
sex	891.0	2	male	-	-	-	-
age	714.0	-	-	29.6991 176470 58	14.5264 973323 34	0.42	80.0
sibsp	891.0	-	-	0.52300 785634 11	1.10274 343229 34	0.0	8.0
parch	891.0	-	-	0.38159 371492 70	0.80605 722112 99	0.0	6.0
fare	891.0	-	-	32.2042 079685 74	49.6934 285971 80	0.0	512.329 2
embarked	889.0	3	S	-	-	-	-
class	891.0	3	Third	-	-	-	-
who	891.0	3	man	-	-	-	-
adult_male	891.0	2	True	-	-	-	-
deck	203.0	7	C	-	-	-	-
embark_town	889.0	3	Southampton	-	-	-	-
alive	891.0	2	no	-	-	-	-

Feature	count	unique	top	mean	std	min	max
alone	891.0	2	True	-	-	-	-

3. Processing & Transformation Logs

Cleaned Dataset Shape: 784 rows x 17 columns

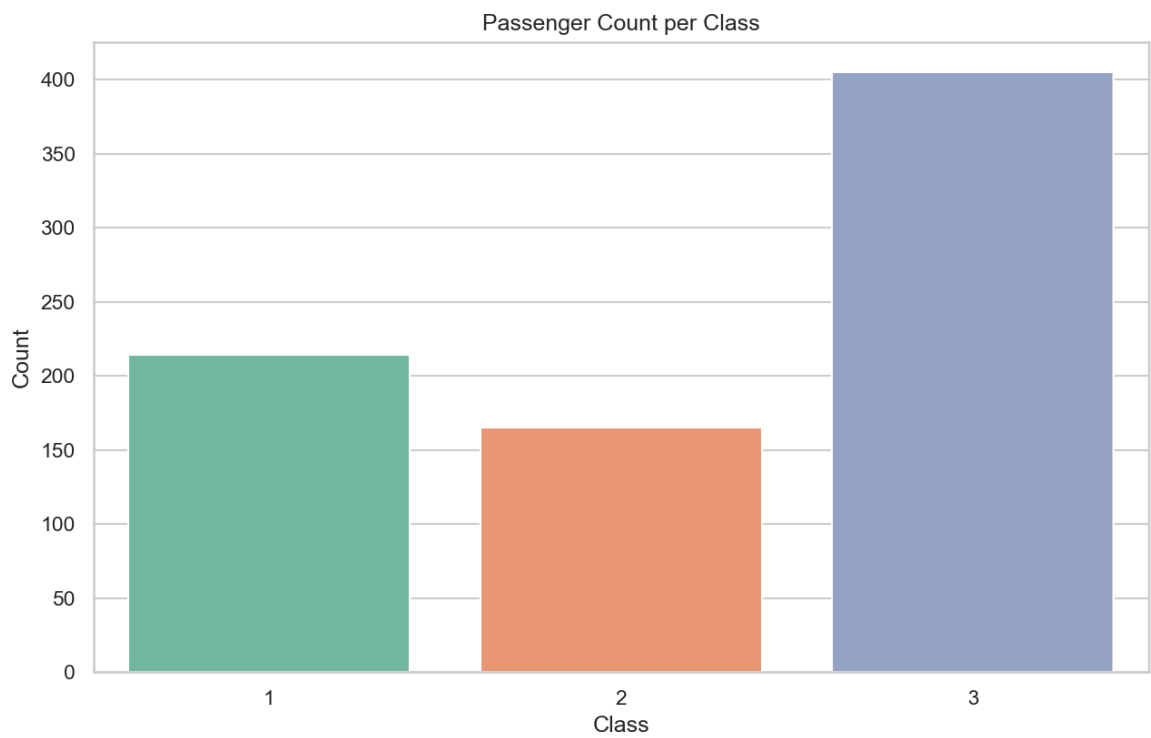
Outlier Handling: IQR method used to cap numerical outliers.

New Features Created: family_size, is_alone

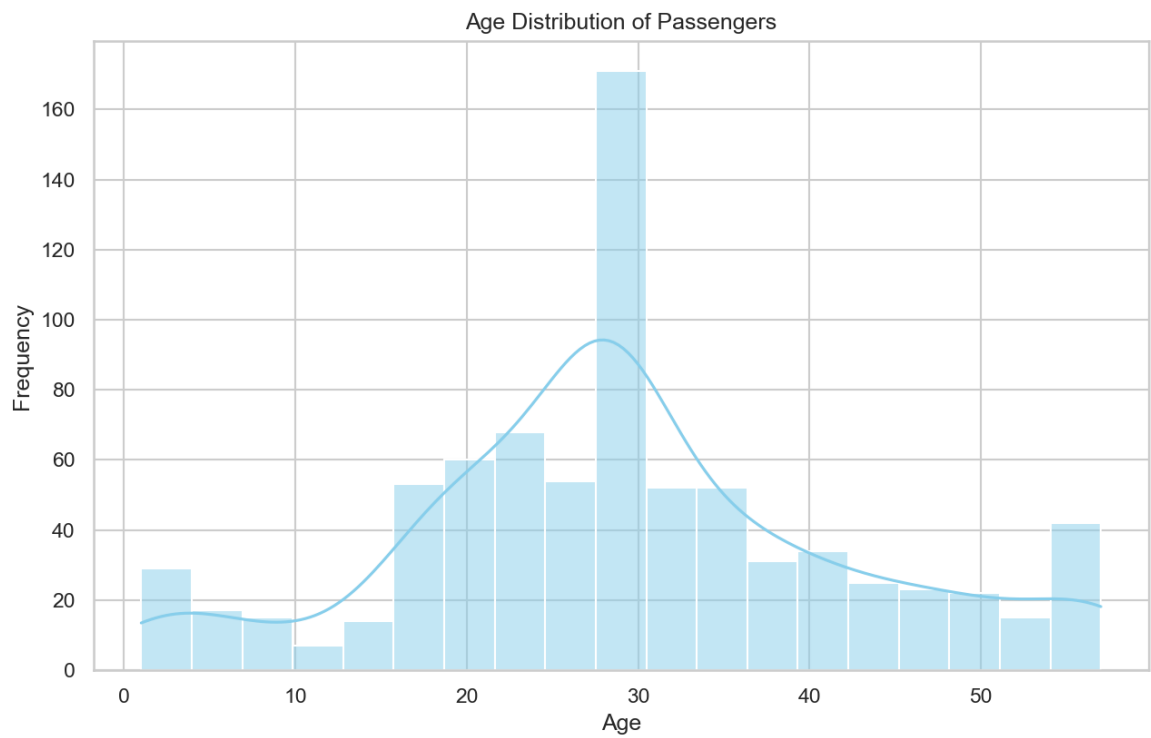
Encoded Categorical Columns: sex, alive

4. Data Visualizations

Bar Passenger Class



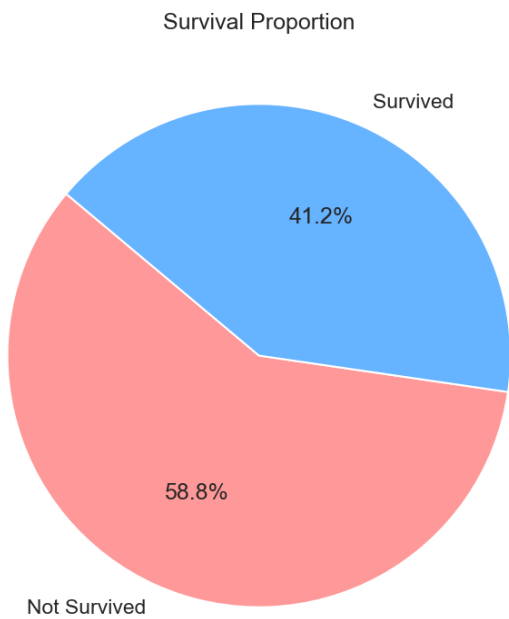
Histogram Age



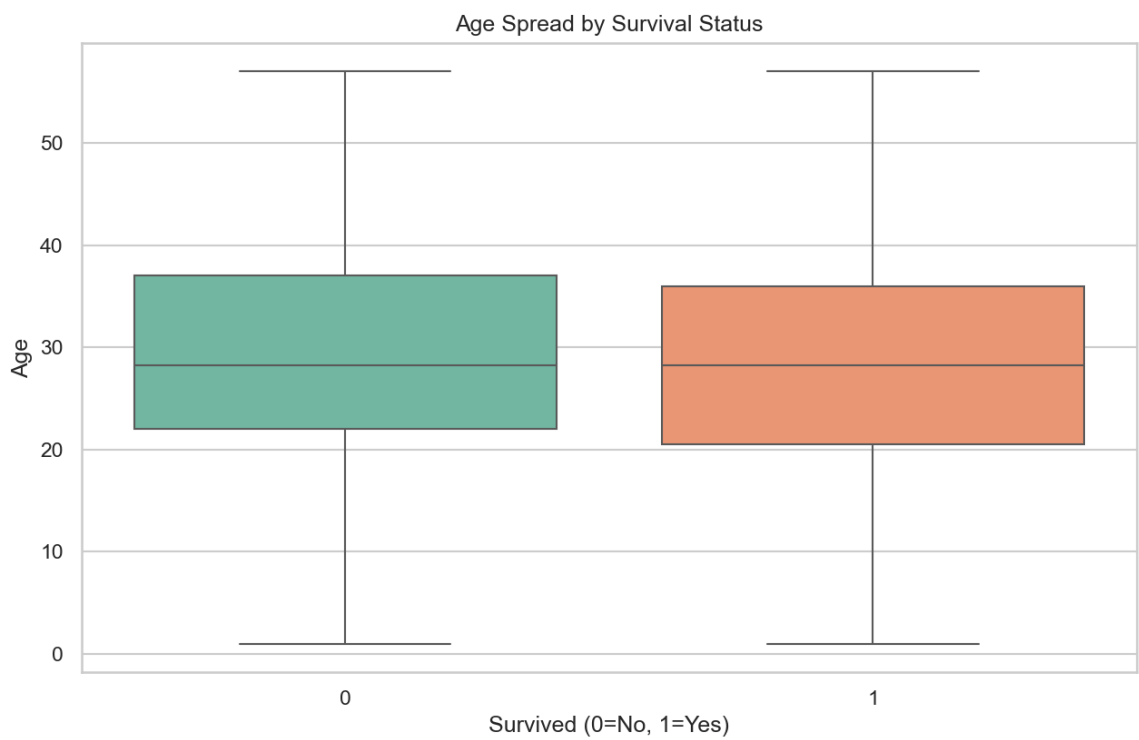
Line Survival Trend



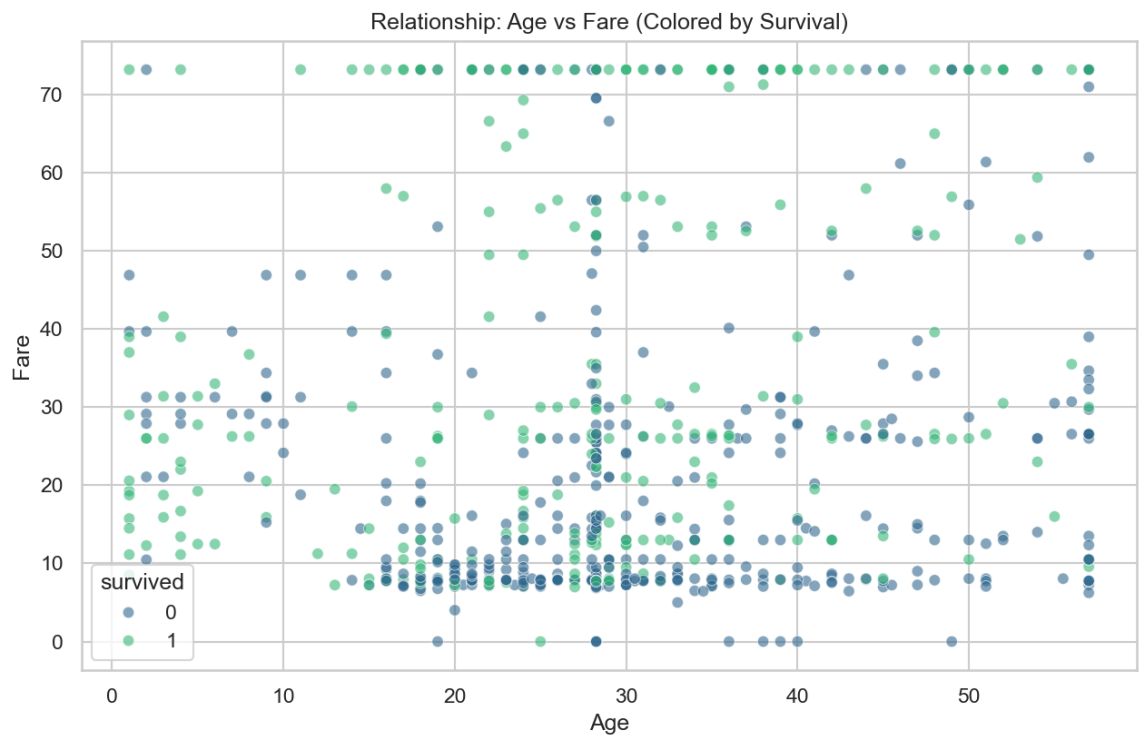
Pie Survival



Boxplot Age Survival



Scatter Age Fare



Heatmap Correlation

